

Bank Nifty Quant Command Center

Client Overview Document

Prepared for: Client Review | Confidential

What Is This Platform?

The **Bank Nifty Quant Command Center** is a personal algorithmic trading assistant for the Indian derivatives market. It is a web-based platform that connects to Angel One SmartAPI to pull live market data, mathematically evaluates Bank Nifty options trading opportunities in real time, and lets you simulate trades safely before risking real capital.

Think of it as a **systematic trading co-pilot** — one that does the analysis, tells you when to act and why, enforces strict risk limits, and keeps a detailed log of every decision made.

The platform is live and accessible at:

■ <https://stock-trading-1-cquo.onrender.com>

The Problem It Solves

Retail traders lose money primarily for three reasons:

1. **Emotional decisions** — buying or selling based on gut feel, news, or panic
2. **Over-trading** — placing too many trades without a clear edge
3. **No audit trail** — not knowing what worked, what didn't, or why

This platform eliminates all three by enforcing a quantitative, rule-based approach to every trade decision.

How It Works — The 3-Layer System

Layer 1 — Live Market Data

The platform connects to **Angel One SmartAPI** and the **NSE India feed** to pull:

- Bank Nifty spot price (live)
- Put-Call Ratio (PCR) data — collected every few minutes throughout the trading day
- Bank Nifty constituent stock prices (HDFCBANK, ICICIBANK, KOTAKBANK, AXISBANK, SBIN)
- Currency exchange rates (USD, EUR, GBP, JPY)
- Full NSE option chain data

Layer 2 — The Signal Engine (3 Algorithms Running Together)

The platform does not guess. It uses three mathematical models simultaneously, and only generates a trade signal when all three agree:

Algorithm 1 — PCR Rolling Z-Score

The Put-Call Ratio measures market sentiment. Instead of fixed thresholds (which fail during trending markets), we compute a 30-day rolling Z-score:

$$Z_{\{PCR\}} = \frac{PCR_t - \mu_{\{30d\}}}{\sigma_{\{30d\}}}$$

A signal is only considered when $Z < -1.2$ (market oversold, CE opportunity) or $Z > +1.2$ (market overbought, PE opportunity).

Algorithm 2 — Fibonacci + Central Pivot Range (CPR)

Bank Nifty's price levels are analysed using:

- The 0.618 Golden Ratio retracement (the most respected support/resistance in technical analysis)
- The Central Pivot Range (CPR) — calculated from the previous day's High, Low, and Close

A trade is only considered when price is near a key level — not in open air.

Algorithm 3 — Weighted Constituent Bank Stock Breadth

Since HDFCBANK (28.5% weight) and ICICIBANK (23.1% weight) together control over 51% of the Bank Nifty index, the platform checks whether the major bank stocks are moving in the same direction as the signal. This prevents buying Call options when HDFCBANK is quietly falling.

The Confluence Rule: All three must agree. A Z-Score signal alone is not enough. A Fibonacci level alone is not enough. All three must confirm simultaneously before any action is recommended.

Layer 3 — Paper Trading Simulation OMS

Before any real money is involved, every signal is executed in a **paper trading environment** — a simulated Order Management System (OMS) that:

- Applies a realistic **0.05% slippage penalty** to simulate real market execution
- Tracks your virtual balance (starting at **₹1,00,000**)
- Records every trade with entry price, exit price, P&L, and the exact reason the signal was generated

Risk Protection Built Into the System

The platform has multiple layers of capital protection that cannot be overridden:

Protection	Rule	Purpose
Max 5 trades per day	No more than 5 paper trades in any single trading day	Prevents over-trading on noisy markets
15-minute cooldown	After every trade, the system pauses for 15 minutes	Allows post-trade volatility to settle before re-entering
Daily loss circuit breaker	Trading freezes if cumulative daily loss exceeds ₹5,000	Hard stop to protect capital
Quarter-Kelly position sizing	Each trade size is calculated using fractional Kelly criterion (25%)	Mathematically optimal bet sizing that prevents ruin
Emergency Kill Switch	One-click button visible at all times in the header	Immediately freezes all signals and paper positions
DSR Gate for live trading	Live real-money execution is locked until the paper trading Deflated Sharpe Ratio (DSR) reaches ≥ 0.95 over 14 days	Ensures the strategy is proven before real capital is used

***The DSR Gate is the most important feature.** It means no live trading can happen until the system has mathematically demonstrated an edge over at least two weeks of forward simulation. This is standard institutional practice.*

What You See — The 6 Tabs

■ Strategy Signals (Tab 1)

The main dashboard. Shows:

- The current active signal: ■ BUY CE / ■ BUY PE / ■ NEUTRAL HOLD
- Signal confidence percentage (currently 92% when triggered)
- The exact rationale — which Fibonacci level was touched, what the Z-score was, what breadth was doing
- Live Fibonacci levels and CPR pivots calculated from the current spot price
- A one-click paper trade execution button

■ Portfolio OMS (Tab 2)

Your paper trading account dashboard. Shows:

- Current virtual balance
- Realised P&L
- Win rate percentage
- Active open positions with entry price, stop-loss, and profit target
- Buttons to execute quick trades and trigger the kill switch

■ Weekly Audit (Tab 3)

A weekend performance review dashboard. Shows:

- Total trades executed over the 7-day period
- Win rate, net P&L, max drawdown
- A complete trade-by-trade log including: timestamp (IST), trade direction (BUY CE / BUY PE), strike price, entry/exit price, P&L, and the exact signal rationale that triggered each trade
- Export to CSV or JSON for offline analysis

■ Strategy Audit (Tab 4)

Full mathematical transparency. Lets you inspect:

- **Algo 1 view:** The current raw PCR value, 30-day mean, standard deviation, and calculated Z-score — the exact numbers feeding the signal
- **Algo 2 view:** Live Fibonacci levels and CPR pivots derived from the current spot price
- **Algo 3 view:** Individual bank stock % changes used in the breadth calculation
- **■ Strategy Masterclass E-Book:** A 6-chapter interactive guide explaining every concept from scratch — options basics, PCR, Fibonacci, Kelly Criterion, and how this platform works — written for someone with no prior trading experience

■ Master Market Grid (Tab 5)

A real-time market overview including:

- NSE Indices: NIFTY, Bank Nifty, SENSEX, INDIA VIX
- Individual Bank Nifty constituent stocks with price, volume, and % change

- Full NSE option chain viewer (select any symbol and expiry)
- PCR historical chart
- Live currency exchange rates (USD/INR, EUR/INR, GBP/INR, JPY/INR)

■ Risk & Settings (Tab 6)

Configure your trading parameters:

- Initial capital allocation per trade
- Max lot size
- Daily max loss cap
- Trailing stop-loss %
- Profit target %
- Access to the interactive client requirements survey

Data Integrity & Transparency

Every number shown in the platform is traceable:

- The signal page shows the **live spot price** the calculation was based on
- The audit page shows the **exact raw inputs** (PCR value, Z-score, breadth %) that produced each signal
- Every paper trade logs **the complete signal rationale** that triggered it — not just "bought CE" but "PCR Z-Score: -1.34 + Price at Fib 0.618 (■57,429) + HDFC/ICICI advancing 73.2%"
- The weekly log persists across server restarts — no data is lost

Platform Status & Live Deployment

Component	Status
Live web deployment	■ Active on Render
Angel One SmartAPI connection	■ Authenticated with credential storage
PCR background collector	■ Runs automatically after login
Paper trading OMS	■ Fully operational
Signal engine	■ Live with 3-way confluence gate
Weekly audit logging	■ Persists to disk, survives restarts
Market data (NSE)	■ No authentication required
Currency rates (NSE)	■ Cached with freshness indicators

Current Stage — Paper Trading Simulation

The platform is currently in its **Paper Trading Validation Phase**. This is intentional and important.

During this phase:

- All trades are simulated — no real money is placed

- Every trade decision is logged with full rationale
- Performance metrics (win rate, Deflated Sharpe Ratio, drawdown) are being tracked over time
- The system is being observed to verify that the edge seen in research holds up in forward real-time conditions

Expected duration: Approximately 14 trading days (3 calendar weeks)

Once the Deflated Sharpe Ratio ($DSR \geq 0.95$) is achieved and sustained, the live execution pathway will be unlocked.

What the Platform Does NOT Do

To set accurate expectations:

- It does **not** guarantee profits. No algorithm can.
 - It does **not** replace human judgment — it informs it.
 - It does **not** execute real trades automatically. Every trade is manually confirmed.
 - It does **not** trade every day — on days with no confluence, it correctly stays in cash.
 - It does **not** use social media sentiment, news, or tips. Every decision is purely mathematical.
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Credentials & Security

- Angel One API credentials are stored securely on the server, not in the browser
 - TOTP-based two-factor authentication is used for every session
 - Credentials are never transmitted in plaintext
 - Session expiry is handled with automatic silent re-authentication
 - The platform is designed for single-user private use — no public registration
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Glossary of Key Terms

Term	Plain English Meaning
PCR (Put-Call Ratio)	A measure of how many people are buying "insurance" vs betting on a rise. High PCR = market fear. Low PCR = market confidence.
Z-Score	How unusual the current PCR reading is compared to the last 30 days. A score below -1.2 means the market is unusually optimistic — a contrarian CE signal.
Fibonacci 0.618	A price level derived from mathematical ratios where markets commonly reverse or bounce. Known as the "Golden Ratio" level.
CPR (Central Pivot Range)	A daily price zone calculated from yesterday's High + Low + Close. Institutional traders watch this closely.
Breadth	Whether the big bank stocks are mostly going up or mostly going down — used to confirm the direction of the signal.
Kelly Criterion	A formula from probability theory that calculates the optimal amount to bet relative to your edge. We use 25% (Quarter-Kelly) for safety.
DSR (Deflated Sharpe Ratio)	A rigorous version of the Sharpe Ratio that accounts for the risk of having tested many strategies and gotten lucky. $DSR \geq 0.95$ is considered statistically verified.
Paper Trading	Simulated trading with virtual money — identical to real trading in every way except no actual money is at risk.
CE / PE	Call option (bet on market rising) / Put option (bet on market falling).
ATM Strike	At-The-Money — the option strike price closest to where the market is right now.
Lot Size	One lot of Bank Nifty options = 15 units. Most positions are 1 lot.

This document reflects the platform as of August 2026. For questions or a live walkthrough, please request a demo session.